Calibration of a Skipper CCD quantum image sensor in a new cryogenic system

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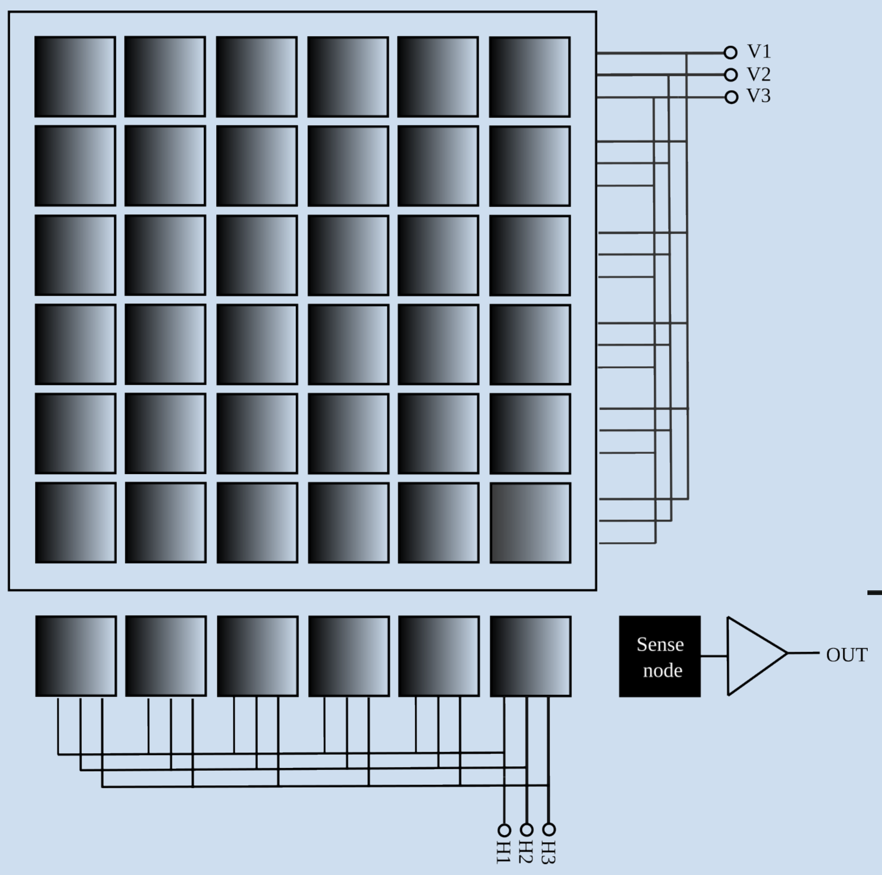
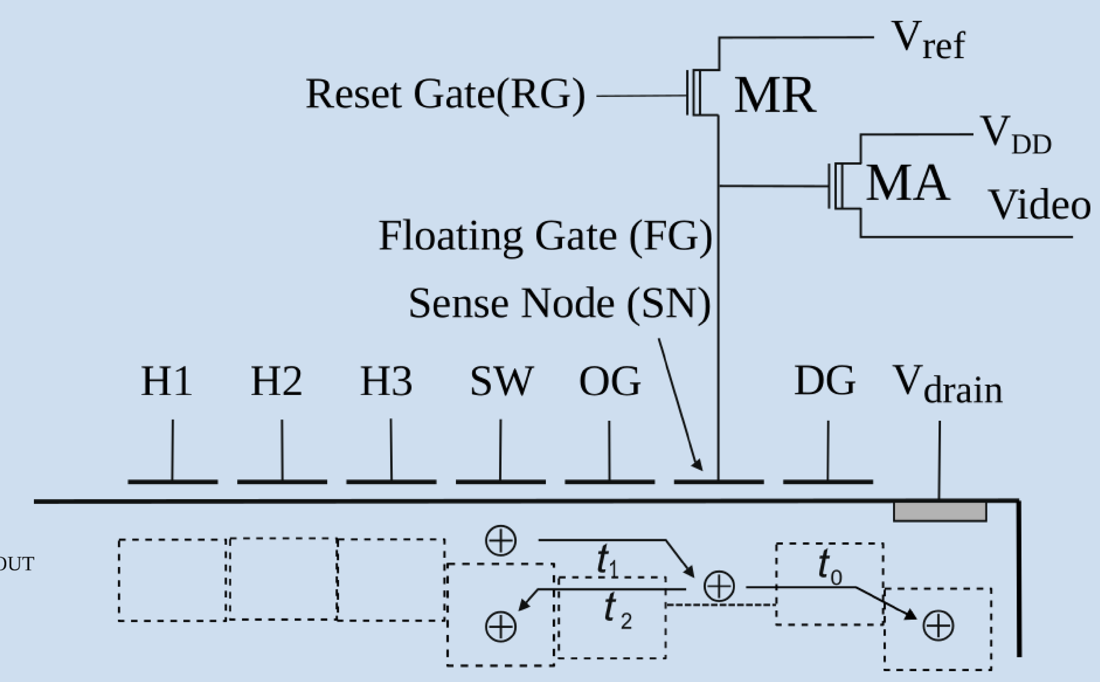
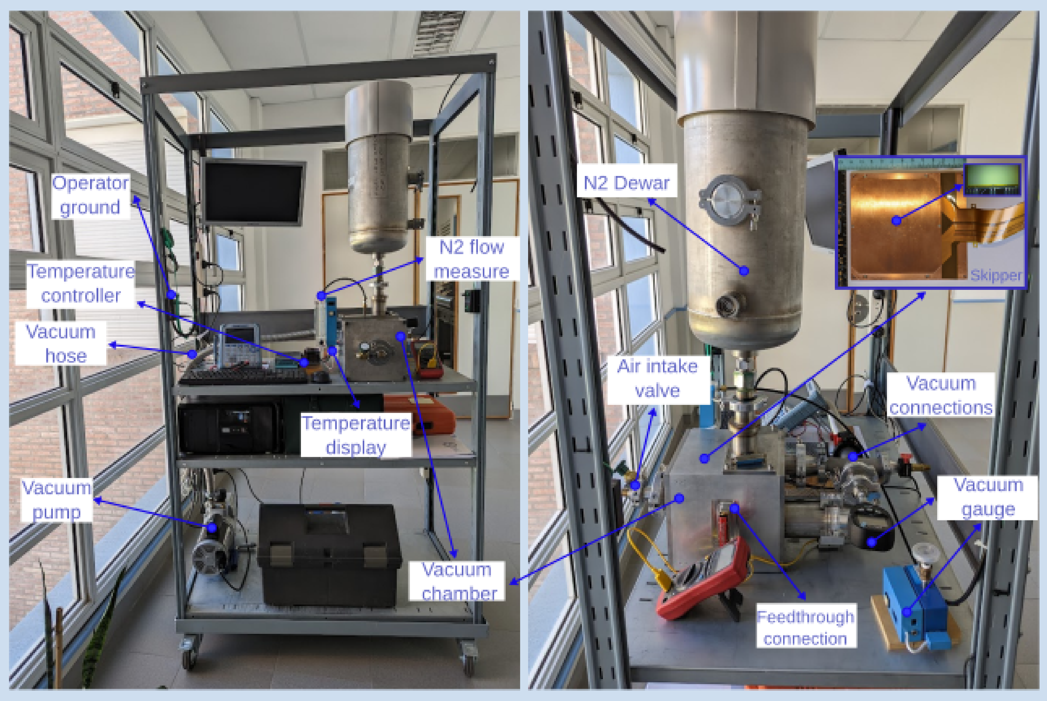
Introduction

A *Charge-Coupled Device (CCD)* is an imaging technology, used across many scientific disciplines. This semiconductor device features an arranged matrix of pixels, each functioning as a capacitor to store charges. The images are composed by shifting the charge and reading the pixels sequentially, using the output stages. The Skipper-CCD has the unique capability of measuring the charge in one pixel several times, thus being able to achieve electron-resolution images. Below, a diagram depicting the functioning of a CCD is shown on the left, a typical readout stage for the Skipper-CCD is in the middle, and the experimental setup, illustrating the cryogenic equipment enabling the sensor to operate at the nominal temperature of 148.15 K, is displayed on the right. This configuration aims to mitigate noise contributions.







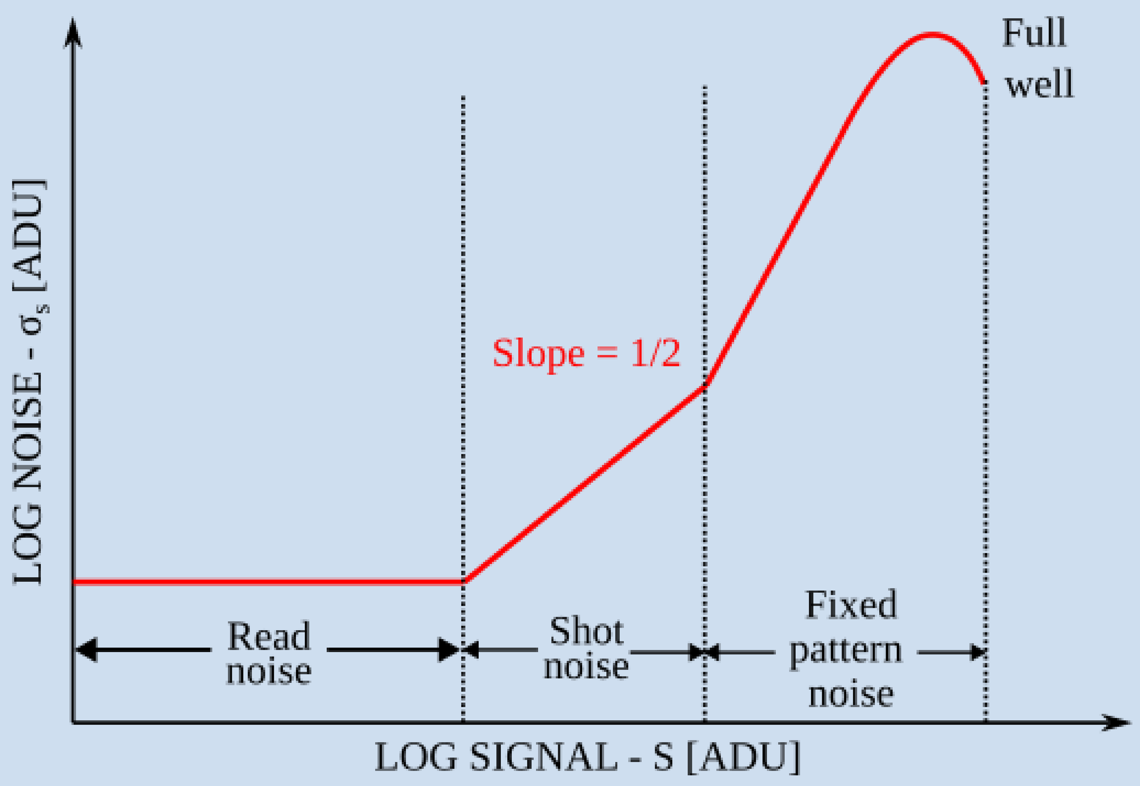
The conversion of incident photons to a measurable output entails a series of transfer functions, culminating in a representation in analog-to-digital units (ADU). Interpreting CCD data necessitates translating these units back into electron counts. This is done through the application of a gain factor K. Two methods for this computation were explored:

- The standard Photon Transfer Curve (PTC),
- A novel approach that uses the ability of Skipper-CCDs to achieve electron-resolution. We call it the Electron Charge Measurement (ECM) method.

Methods

Photon Transfer Curve

The PTC is derived by illuminating the sensor at different light intensities and then plotting the standard deviation against the mean signal. The plot reveals three distinct noise regions.



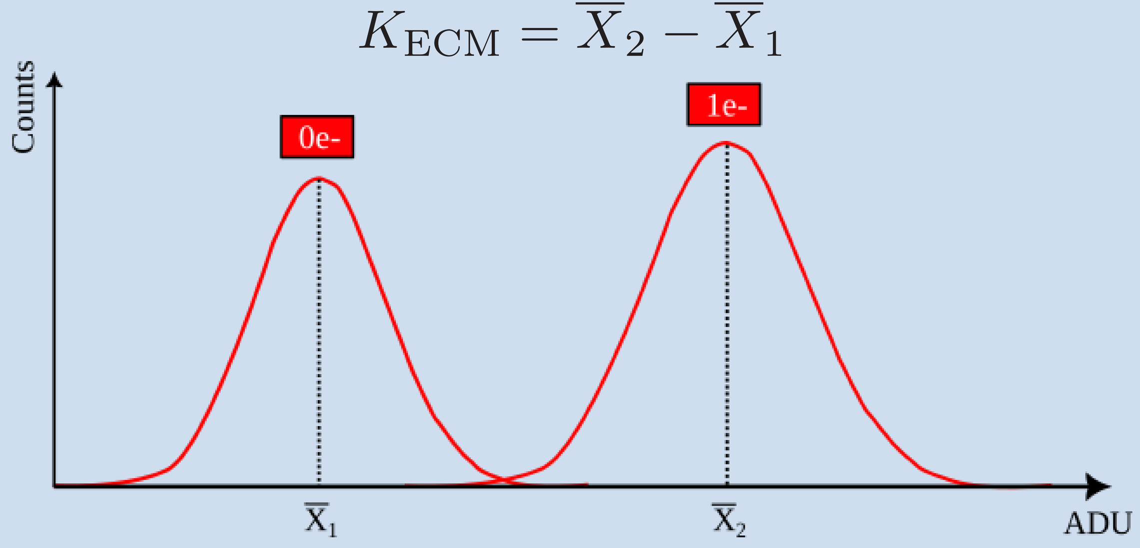
As the illumination intensifies, the noise gradually becomes dominated by shot noise due to the nature of photon arrivals. Then, gain is derived as:

$$K_{\mathrm{PTC}} = \frac{S[\mathrm{ADU}]}{\sigma_s^2[\mathrm{ADU}]^2 - \sigma_{\mathrm{read}}^2[\mathrm{ADU}]^2} = \frac{S[\mathrm{ADU}]}{\sigma_{\mathrm{shot}}^2[\mathrm{ADU}]^2}$$

Electron Charge Measurement

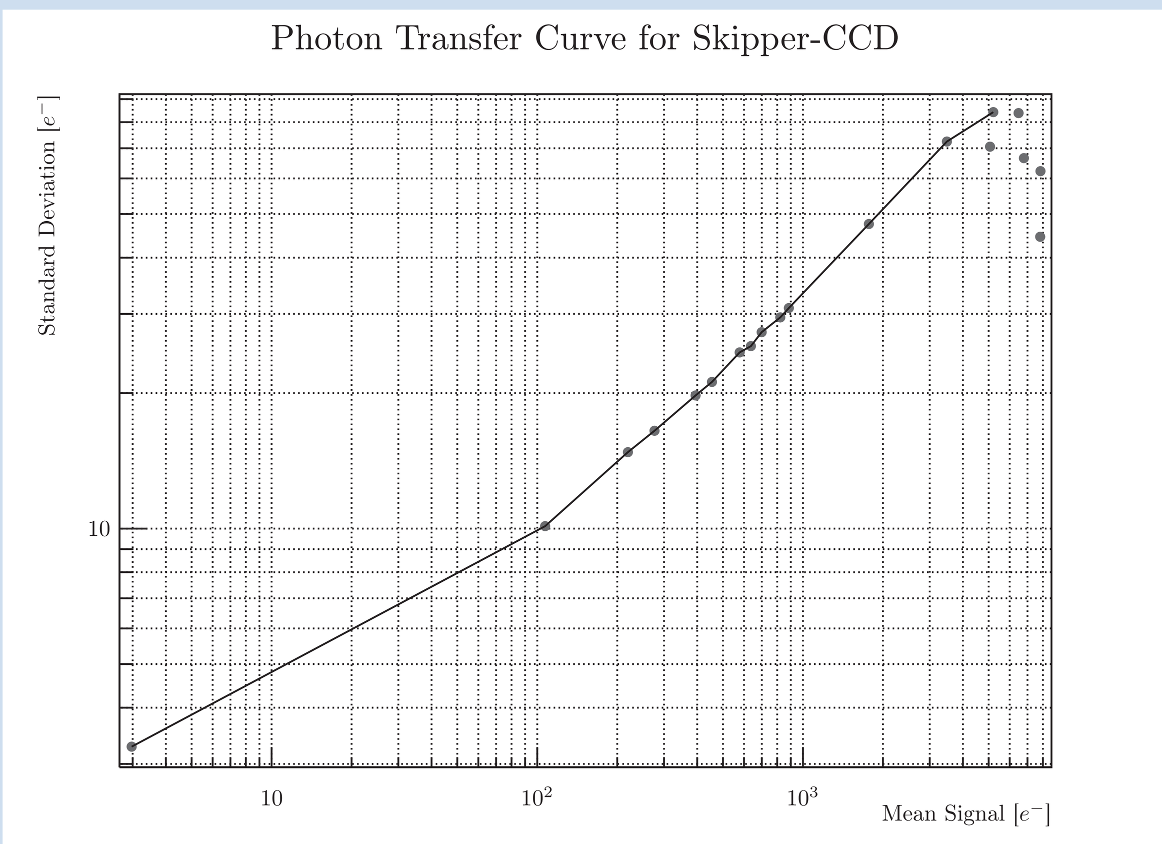
In the image histogram, distinct Gaussian distributions in ADU represent individual charge packets. The gain is inferred by fitting adjacent Gaussian curves and calculating the difference between their means:

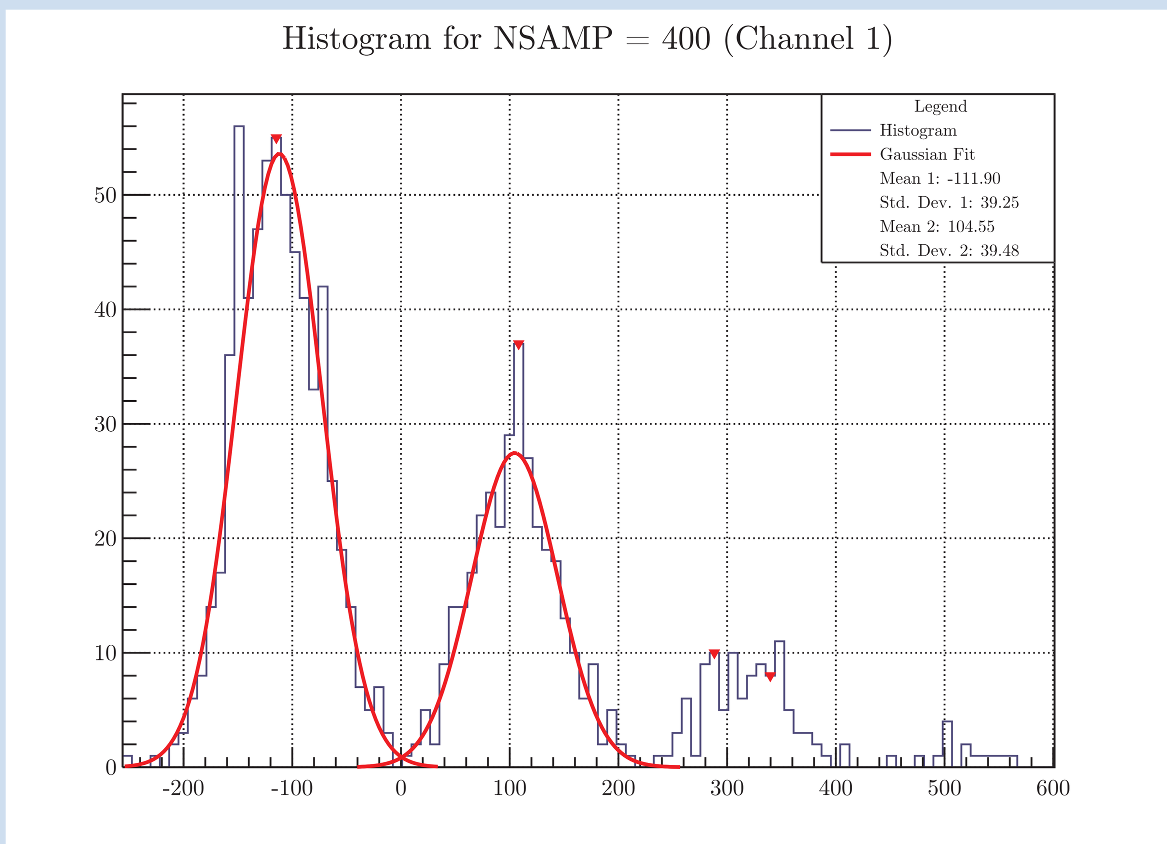
$$K_{\mathrm{ECM}} = \overline{X}_2 - \overline{X}_1$$



Results

- To calculate the PTC, 19 images were taken; each pixel was sampled once (NSAMP = 1). The readout noise for this set was approximately $\sigma_{\mathrm{read}} \approx 3e^-$. The PTC is depicted on the left side below.
- For images with single electron resolution, NSAMP = 400 samples per pixel were used, yielding a readout noise of approximately $\sigma_{\mathrm{read}} \approx 0.18e^-$. The histogram corresponding to the ECM method is displayed on the right side below.





The calculated gains for three channels, based on the analysis of the figures above, are detailed below:

Channel	K_{PTC} [ADU/e^-]	K_{ECM} [ADU/e^-]	Difference
1	241.98	216.45	10.55%
2	236.89	211.22	10.84%
3	248.66	226.70	8.83%

Discussion

- The percentage difference exceeding 8% could be partially attributed to sensor non-linearity, because K_{ECM} was measured using 0 and 1 electron and K_{PTC} using hundreds to thousands.
- The choice of calibration method depends on the application and available resources. If the objective is to achieve a sub-electron noise regime, the gain could be calculated using the Gaussian fitting.

Conclusions

This study compared two methods for gain characterization in Skipper CCDs. While the PTC method is generally reliable, it is less precise for sub-electron noise levels. In contrast, the latter method excels in high-precision applications but comes at the expense of increased computation time.